Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form	
Lab Name	**OSI Model, TCP/IP Model and Basic LAN Implementation**
Lab Number	**04**
Submitted by:	
Student Names	**Registration #**
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Rubrics		**Marks**		
		1	**2**	**3**
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab Tasks

Post Lab Task 1:

Implement a basic LAN network between two computers using Ethernet cable and explain all steps in detail. Draw a flow chart also to explain all steps.

Objective:

- Name the seven layers of the OSI model in order using a mnemonic.
- Describe characteristics, functions, and keywords for each OSI layer.
- Describe the encapsulation units for each layer.
- Identify physical devices/components that operate at each layer.
- Explain the four layers of the TCP/IP model.
- Map OSI layers to TCP/IP layers.
- List primary protocols and utilities for each TCP/IP layer.
- Implement a basic LAN network between two computers using Ethernet.

Lab Steps:

Step 1: List the OSI Model Layers with Mnemonic and Functions:

Layer	Name	Mnemonic	Function/Keywords
7	Application	All	User interface, network services.
6	Presentation	People	Data format, encryption, compression.
5	Session	Seem	Sessions, dialog control, sync.
4	Transport	To	Reliability, flow control, segmentation.
3	Network	Need	Routing, addressing (IP), packet forwarding.
2	Data Link	Data	MAC addressing, error detection, frame synchronization.
1	Physical	Processing	Cables, signals, binary transmission.

Mnemonic: **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing.

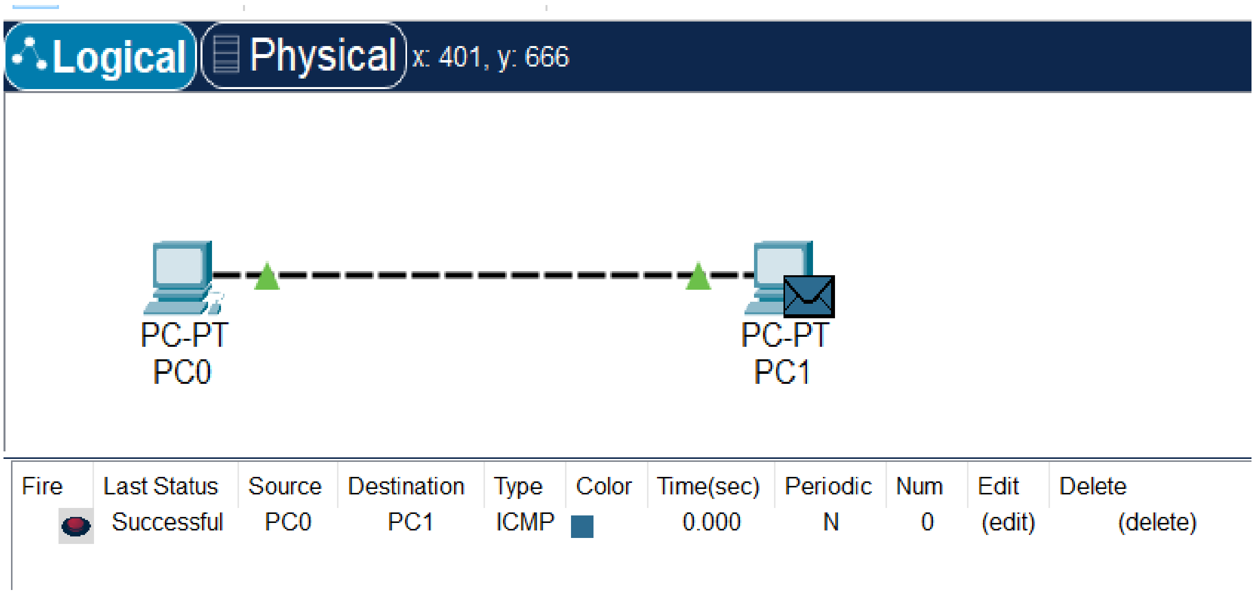
Step 2: OSI Layers with Encapsulation Units and Devices:

Layer	Encapsulation Unit	Device
Application	Data	Computer, software apps.
Presentation	Data	Gateways, translators.
Session	Data	Gateways, session software.
Transport	Segment	Firewall
Network	Packet	Router
Data Link	Frame	Switch, NIC (MAC Address).
Physical	Bits	Hub, Cable, Repeater, NIC.

Step 3: OSI vs TCP/IP Model Comparison Table:

OSI Layer No.	OSI Layer Name	TCP/IP Layer No.	TCP/IP Layer Name	Encapsulation Unit	Protocols/Utilities
7	Application	4	Application	Data	HTTP, FTP, DNS, SMTP, Telnet, SSH
6	Presentation	4	Application	Data	SSL, TLS, JPEG, MPEG
5	Session	4	Application	Data	NetBIOS, RPC
4	Transport	3	Transport	Segment	TCP, UDP
3	Network	2	Internet	Packet	IP, ICMP, ARP, IGMP
2	Data Link	1	Network Interface	Frame	Ethernet, PPP, MAC, Switch protocols
1	Physical	1	Network Interface	Bits	Cables, Hubs, NIC, Modems

Implementing a LAN Network:



Steps to Setup LAN:

1. **Connect Ethernet Cable:**
   - Plug one end of the Ethernet cable into each computer's Ethernet port.

2. **Turn Off Firewall (or configure to allow sharing):**
   - Go to Control Panel → System and Security → Windows Firewall.
   - Choose "Turn Windows Firewall On or Off" → Turn Off for Private network.

3. **Assign IP Addresses:**
   - Go to Network and Sharing Center → Change adapter settings.
   - Right-click Ethernet → Properties → Select "Internet Protocol Version 4 (TCP/IPv4)" → Properties.
   - Assign static IPs (e.g., PC1: 192.168.1.1, Subnet: 255.255.255.0; PC2: 192.168.1.2, same subnet).

4. **Create a Homegroup (Windows 7/8):**
   - Go to Control Panel → Home Group → Create a new group.

5. **Join the Homegroup:**
   - On the second computer, go to Home Group settings → Join the created Home Group using the password.

6. **Share Drives/Folders**
   - Right-click on any folder/drive → Properties → Sharing tab → Share.

7. Verify Connection

- Ping one computer from the other: ping 192.168.1.2 from PC1 and vice versa.

Flowchart: LAN Implementation Steps: